

THE ORTHOGONAL POLYNOMIALS OF THE TRUNCATED POISSON PROBABILITY DISTRIBUTION

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The truncated Poisson probability distribution is given by

$$(1) \quad P_x = \frac{e^{-\lambda} \lambda^x}{x! (1 - e^{-\lambda})}, \quad x=1,2,\dots$$

$$= \frac{\lambda^x}{x! (e^{\lambda} - 1)}$$

The R.H.S. of (1) is the probability distribution of a Noack series (1950) $P_x = A_x Z^x / F(Z)$ in which $Z = \lambda$ and

$$\sum_x A_x Z^x = F(Z) = F(\lambda) = e^{\lambda} - 1 = F \text{ (say)}$$

Also

$$A_x = \frac{1}{x!} > 0 \quad (x=1,2,\dots)$$

Van Heerden (1961) showed that the r -th factorial moment of this discrete power series probability distribution is given by

$$\mu'_r = U_r = \sum x^{(r)} A_x Z^x / F(Z)$$

$$= Z^r \frac{\partial^r F(Z)}{\partial Z^r} / F(Z)$$

where $x^{(r)} = x(x-1)(x-2) \dots (x-r+1)$.

Hence the r -th factorial moment of P_x is given by

$$U_r = \lambda^r \frac{\partial^r (e^{\lambda} - 1)}{\partial^r \lambda} / (e^{\lambda} - 1)$$

$$= \lambda^r \frac{F+1}{F} \text{ (where } F = e^{\lambda} - 1)$$

The orthogonal polynomials up to the third degree of the power series probability distribution have been derived by Van Heerden and Gonin (1966). We use the notation given in that paper.

We have

$$G_0(x) = 1$$

$$\begin{aligned}
 G_1(x) &= x - U_1 = x - \lambda \frac{F+1}{F} \\
 U'_r &= \frac{\lambda^r e^\lambda}{e^\lambda - 1} + \frac{r \lambda^{r-1} e^\lambda}{e^\lambda - 1} - \frac{\lambda^r e^\lambda e^\lambda}{(e^\lambda - 1)^2} \\
 &= \frac{F+1}{F^2} [rF - \lambda] \lambda^{r-1}
 \end{aligned}$$

Hence

$$\frac{A_{21}}{A_{22}} = -\frac{U'_2}{U'_1} = -\frac{(2F - \lambda)}{(F - \lambda)}$$

and

$$\frac{A_{20}}{A_{22}} = \frac{U_1 U'_2 - U_2 U'_1}{U'_1} = \frac{\lambda^2 (F+1)}{(F - \lambda)}$$

Hence

$$G_2(x) = x^{(2)} - \frac{\lambda(2F - \lambda)x}{(F - \lambda)} + \frac{\lambda^2(F+1)}{(F - \lambda)}$$

We find

$$\begin{aligned}
 A_{33} &= \frac{(F+1)^2}{F^3} [2F - \lambda(\lambda + 2)] \\
 A_{32} &= -\frac{(F+1)^2 \lambda}{F^3} [6F - 2\lambda(\lambda + 3)] \\
 A_{31} &= \frac{(F+1)^2 \lambda^2}{F^3} [6F - \lambda(\lambda + 4)] \\
 A_{30} &= \frac{-2\lambda^3(F+1)}{F^3}
 \end{aligned}$$

Hence

$$\begin{aligned}
 G_3(x) &= x^{(3)} - \frac{\lambda[6F - 2\lambda(\lambda + 3)]x^{(2)}}{[2F - \lambda(\lambda + 2)]} \\
 &\quad + \frac{\lambda^2[6F - \lambda(\lambda + 4)]x}{[2F - \lambda(\lambda + 2)]} - \frac{2\lambda^3(F+1)}{[2F - \lambda(\lambda + 2)]}
 \end{aligned}$$

The following Table A compiled by Steyn and Gerntholtz (1968) sets out the fitting of a set of data θ_x to a truncated Poisson probability function

$$P_x = \frac{\lambda^x}{x! (e^\lambda - 1)}, \quad (x \geq 1)$$

It is estimated from the expression $\frac{\lambda}{\bar{x}} = 1 - e^{-\lambda}$. It can easily be shown that the estimator $\hat{\lambda}$ of λ obtained from this equation is a maximum likelihood estimator.

$$\text{Also } \bar{x} = \frac{\lambda}{1 - e^{-\lambda}} \text{ and } F = \frac{\lambda}{\bar{x} - \lambda}$$

TRUNCATED POISSON PROBABILITY DISTRIBUTION

TABLE A
UNIVERSITY OF PRETORIA – EUROPEAN STUDENTS
 (Time – during examination)

x (No. in group)	Observed No. of groups θ_x	Expected No. of groups	Value of χ^2
1	751	738.7	0.2
2	352	362.0	0.3
3	110	118.3	0.6
4	32	29.0	0.3
5	8	5.7	1.6
6	2	1.0	
	1255	1254.7	3.0

$$\hat{\lambda}=0.98$$

$$\bar{x}=\hat{\mu}=1.566$$

$$p>30\% \\ <50\%$$

Steyn and Gerntholtz used the following formulas for fitting the data

$$P_0 = \frac{1}{e^{\hat{\lambda}} - 1} = k \text{ (say)}$$

$$P_1 = \frac{\hat{\lambda}}{e^{\hat{\lambda}} - 1} = k \times \hat{\lambda}$$

$$P_2 = \frac{\hat{\lambda}^2}{2(e^{\hat{\lambda}} - 1)} = k \times \hat{\lambda} \times \frac{\hat{\lambda}}{2}$$

The expected frequencies are then obtained by multiplying throughout by n.
 Let us fit the data θ_x to the curve

$$y_x = n [1 + a_1 G_1(x) + a_2 G_2(x)] P_x$$

by Aitken's method. The coefficient a_1 becomes zero (see H.T. Gonin 1961, p. 120).
 From the data above

$$\lambda = 0.98 \quad F = \frac{.98}{1.566 - .98} = 1.67.$$

Inserting these values in the formula for $G_2(x)$ we obtain

$$G_2(x) = x^{(2)} - 3.35x + 3.71.$$

Let us choose a_2 to be such that

$$y_1 = n [1 + a_2 (x^{(2)} - 3.35x + 3.71)] P_x = 751$$

where 751 is the observed data at $x=1$. This method has often been followed by research workers and implies the insertion of one restraint. It is a convenient deviation from the usual Aitken method.

$$\text{Hence } 1255 [1 + .36 a_2] P_x = 751$$

but

$$1255 P_x = 738.7$$

hence

$$1 + .36 a_2 = \frac{751}{738.7} = 1.01665$$

hence

$$a_2 = .046$$

$$y_2 = [1 - .046 \times .99] 362 = 345.5$$

$$y_3 = [1 - .046 \times .34] 118.3 = 116.5$$

$$y_4 = [1 + .046 \times 2.31] 29.0 = 32.1$$

$$y_5 = [1 + .046 \times 6.96] 5.7 = 7.5$$

$$y_6 = [1 + .046 \times 13.61] 1 = 1.6.$$

Hence we obtain

TABLE B
UNIVERSITY OF PRETORIA – EUROPEAN STUDENTS
(Time – during examination)

x (No. in group)	Observed No. of groups θ_x	Expected No. of groups	Value of χ^2
1	751	751	0.0
2	352	345.5	0.1
3	110	116.5	0.4
4	32	32.1	0.0
5	8	7.5	0.1
6	2	1.6	
	1255	1254.2	0.5

$p > 70\%$
 $< 80\%$

This represents a much better fit.

This method is analogous to the fitting of data by the Type B probability functions (See Aitken, p. 66).

We have used the probability function $p(x) = [1 + a_2 G_2(x) + a_3 G_3(x)] P_x$ where P_x is the truncated Poisson probability function and $G_r(x)$ is the r -th polynomial, orthogonal to the truncated Poisson probability function. The truncated Poisson probability function by itself does not give such a good fit in this case as the assumptions given by Steyn and Gerntholtz in the derivation of their model is not strictly correct owing to the relatively small population size.

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